

KERALAMMATCH

COMPLETE SYSTEM FLOWCHARTS & ARCHITECTURE SPECIFICATION

Master User & Admin Journey · Ephemeral Reveal · Razorpay Webhook · Admin Governance

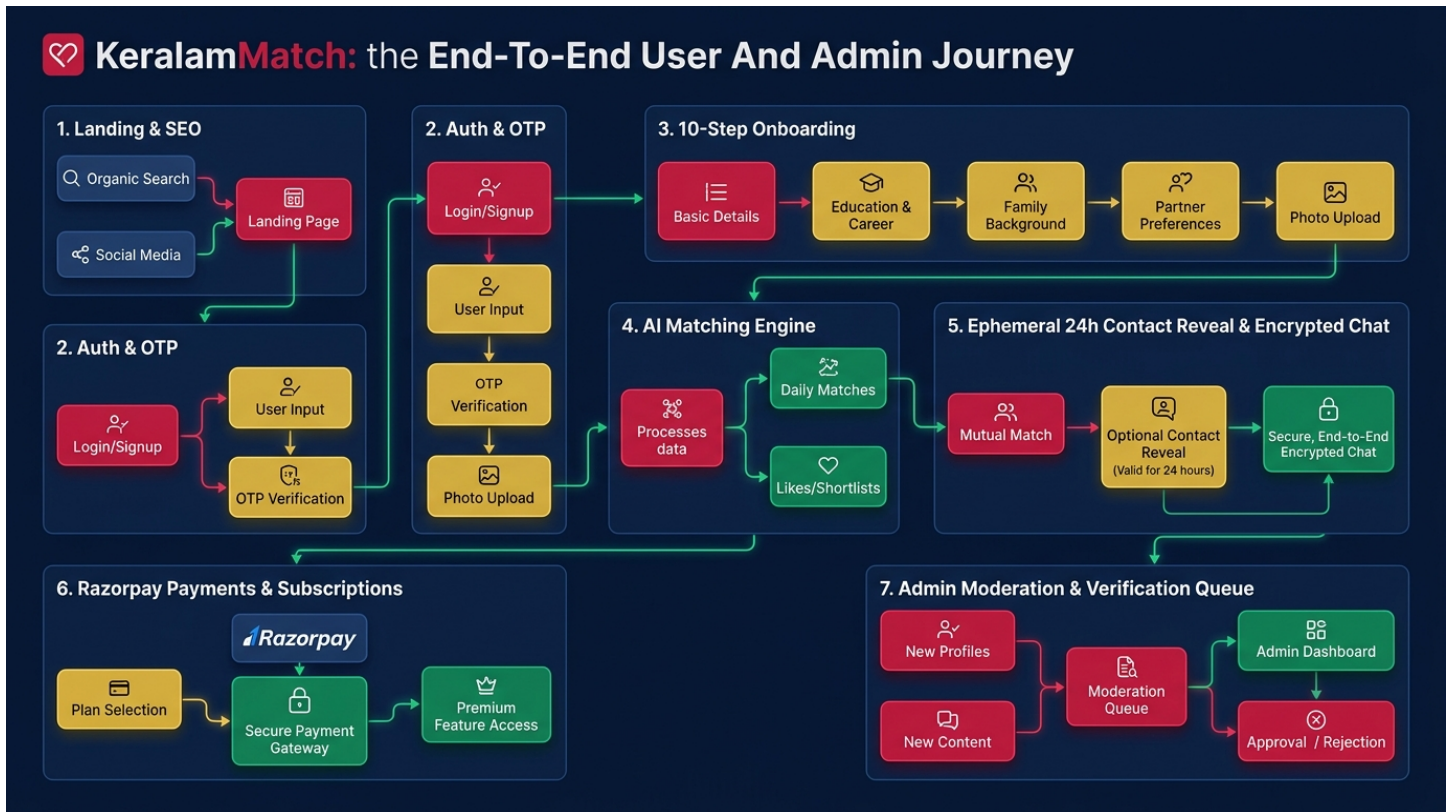


Figure 1: KeralamMatch Complete System Flowchart Visual Overview

1. Master End-to-End User & Admin Journey Flowchart

```

```mermaid
flowchart TD
 %% Styling Configuration
 classDef startEnd fill:#0A1F44,color:#FFFFFF,stroke:#C81D45,stroke-width:2px;
 classDef process fill:#FFFFFF,color:#0A1F44,stroke:#0A1F44,stroke-width:1.5px;
 classDef decision fill:#FCF8F7,color:#C81D45,stroke:#C81D45,stroke-width:2px;
 classDef admin fill:#0A1F44,color:#D4AF37,stroke:#D4AF37,stroke-width:1.5px;
 classDef success fill:#E6F4EA,color:#137333,stroke:#137333,stroke-width:1.5px;

 %% Entry Points & Public Discovery
 A["■ Visitor Discovery
(Landing / District SEO / Community / Blog)"]:::startEnd --> B{"Authenticated Session?"}:::decision

 %% Authentication & Onboarding
 B -- No --> C["■ Auth Page (/auth)
Mobile SMS OTP / Google Sign-In"]:::process
 C --> D{"OTP Verified & Session Set?"}:::decision
 D -- No --> C
 D -- Yes --> E{"Profile Completed?"}:::decision

 %% Onboarding Path
 E -- No --> F["■ 10-Step Onboarding Wizard (/join)
Basic, Religion, Career, Photos, Liveness"]:::process
 F --> G["■ Verification Queue Submit
(Selfie & ID Document Review)"]:::process
 G --> H["■ Member Dashboard (/dashboard)"]:::success

 E -- Yes --> H
 B -- Yes --> H

 %% Core Member Discovery & Matching
 H --> I["■ Discover Matches (/find)
AI Prompt / 12-Attribute Filters"]:::process
 I --> J["■ View Target Profile (/profile/[id])"]:::process

 J --> K{"Action Selected"}:::decision
 K -- Shortlist --> L["■ Added to Saved Shortlist"]:::process
 K -- Direct Message --> M["■ Encrypted In-App Chat (/chat)"]:::process
 K -- Request Contact --> N["■ Send Contact Request (/requests)"]:::process

 %% 24-Hour Ephemeral Reveal Lifecycle
 N --> O{"Recipient Consents?"}:::decision
 O -- Declined --> P["■ Request Closed"]:::process
 O -- Accepted --> Q["■ 24-Hour Ephemeral Window
(AES-256 Decrypted Phone & Email Unlocked)"]:::success
 Q --> Q1{"Timer Expired (>24h)?"}:::decision
 Q1 -- Yes --> Q2["■ Credentials Auto-Lock & Re-mask"]:::process

 %% Monetization & Payments
 H --> R["■ Pricing & Upgrades (/pricing)"]:::process
 R --> S{"Select Purchase"}:::decision
 S -- Membership Tier --> T["■ Razorpay Checkout
(Silver / Gold / Platinum)"]:::process
 S -- Reveal Credits --> U["■ Wallet Refill
(Pay-Per-Reveal Packs)"]:::process
 T & U --> V["■ Webhook Signature Verified & Benefits Provisioned"]:::success

 %% Admin Portal Suite
 W["■ Admin Login (/admin/login)"]:::admin --> X["■ Admin Dashboard (/admin)"]:::admin
 X --> Y1["■ User Management (/admin/users)"]:::admin
 X --> Y2["■ Verification Queue Review (/admin/verification)"]:::admin
 X --> Y3["■ Report Moderation (/admin/reports)"]:::admin
 X --> Y4["■ Financial Operations (/admin/payments)"]:::admin
 X --> Y5["■ Audit Logs & Settings (/admin/audit)"]:::admin
```

```

Operational Flow Breakdown:

- **Visitor Discovery:** Organic visitors discover KeralamMatch via landing page (/), 13 district SEO pages, or 6 community hubs.
- **Authentication & Onboarding:** Mobile SMS OTP (+91) or Google OAuth sets HTTP-only session cookie (km_session), leading to 10-step onboarding (/join).
- **Matching & Profile Viewing:** Algorithmic 12-attribute compatibility scoring (%) on /find and rich profile view on /profile/[id].
- **Action Choices:** Shortlist profile, initiate encrypted in-app chat (/chat), or request 24h contact reveal (/requests).
- **Monetization & Upgrades:** Silver, Gold, Platinum plans and pay-per-reveal credit packs via Razorpay checkout (/pricing).
- **Admin Operations:** Central governance covering user management, verification queue, report moderation, and audit logs.

2. The 24-Hour Ephemeral Contact Reveal Flowchart

```

sequenceDiagram
    autonumber
    actor UserA as Candidate A (Requester)
    participant System as KeralamMatch Backend
    actor UserB as Candidate B (Recipient)
    participant DB as PostgreSQL Database

    UserA->>System: Clicks "Send Contact Request"
    System->>DB: Creates ContactRequest (status: PENDING)
    System->>UserB: Sends In-App Notification & Resend Email
    UserB->>System: Inspects Candidate A Profile & Clicks "Accept"
    System->>DB: Sets status = ACCEPTED, expiresAt = now + 24h
    System->>UserA: Notifies "Request Accepted! 24h Window Active"
    UserA->>System: Opens Profile B (/profile/[id])
    System->>DB: Checks active reveal (now < expiresAt)
    System->>UserA: Decrypts AES-256 Phone & Email + Displays 24:00:00 Countdown
    Note over UserA,UserB: Exactly 24 Hours Pass (now > expiresAt)
    UserA->>System: Requests Profile B contact details
    System-->>UserA: 403 Forbidden: ACCESS_EXPIRED_OR_LOCKED (Auto-locked)

```

Operational Flow Breakdown:

- **Step 1: Request Initiation:** Candidate A clicks 'Send Contact Request' on Profile B. Status set to PENDING.
- **Step 2: Notification & Consent:** Candidate B receives real-time in-app alert and transactional email with profile snippet.
- **Step 3: Mutual Consent Approval:** Candidate B clicks 'Accept'. Backend sets expiresAt = now + 24 hours.
- **Step 4: Decrypted Unlocking:** Candidate A views Profile B. AES-256 decrypted phone & email are displayed with live countdown timer.
- **Step 5: Automatic Re-Locking:** When now > expiresAt, credentials auto-lock and throw 403 ACCESS_EXPIRED_OR_LOCKED.

3. Razorpay Payment & Webhook Architecture Flowchart

```

sequenceDiagram
    autonumber
    actor User as Member
    participant Client as Next.js 16 Client
    participant Server as Next.js Server Action
    participant RZP as Razorpay Gateway API
    participant Webhook as Webhook Route Handler (/api/payments/razorpay/webhook)
    participant DB as PostgreSQL Database

    User->>Client: Selects Silver / Gold / Platinum / Credit Pack
    Client->>Server: Calls checkoutSubscriptionAction(planId)
    Server->>RZP: Creates Order (Amount, INR, Receipt ID)
    RZP-->>Server: Returns Order ID (rzp_order_...)
    Server-->>Client: Initializes Razorpay Standard Modal
    User->>RZP: Completes Payment (UPI / Netbanking / Card)
    RZP-->>Client: Triggers onSuccess callback
    RZP->>Webhook: Dispatches Webhook Event (payment.captured)
    Webhook->>Webhook: Validates HMAC SHA-256 Webhook Signature
    Webhook->>DB: Activates Subscription / Credits Wallet Balance
    Client->>User: Redirects to /pricing/success with Unlocked Features

```

Operational Flow Breakdown:

- **Step 1: Plan Selection:** User selects Silver/Gold/Platinum tier or ₹99 reveal credit pack on /pricing.
- **Step 2: Order Creation:** Next.js Server Action calls Razorpay Order API and returns rzp_order_id.
- **Step 3: Client Checkout:** Standard Razorpay modal opens supporting UPI (GPay/PhonePe), Netbanking, and Cards.
- **Step 4: Webhook Event Dispatch:** Razorpay fires asynchronous webhook event (payment.captured) to /api/payments/razorpay/webhook.
- **Step 5: Signature Validation & Provisioning:** Server validates HMAC SHA-256 signature, updates database status, and provisions membership tier / wallet balance.

4. Admin Governance & Verification Flowchart

```

```mermaid
graph LR
 classDef admin fill:#0A1F44,color:#D4AF37,stroke:#D4AF37,stroke-width:1.5px;
 classDef process fill:#FFFFFF,color:#0A1F44,stroke:#0A1F44,stroke-width:1.5px;
 classDef badge fill:#E6F4EA,color:#137333,stroke:#137333,stroke-width:1.5px;

 A["■ Admin Login (/admin/login)"]:::admin --> B["■ Dashboard Overview (/admin)"]:::admin

 B --> C1["■ Verification Queue (/admin/verification)"]:::admin
 C1 --> C1A["Inspect Selfie Liveness & Aadhaar ID"]:::process
 C1A --> C1B["Approve ■■■ Green Verified Badge Issued"]:::badge
 C1A --> C1C["Reject ■■■ Re-upload Prompt Sent"]:::process

 B --> C2["■ Report Moderation (/admin/reports)"]:::admin
 C2 --> C2A["Review Reported Accounts & Flags"]:::process
 C2A --> C2B["Dismiss / Issue Warning / Suspend User"]:::process

 B --> C3["■ Revenue & Payments (/admin/payments)"]:::admin
 C3 --> C3A["Audit Ledger, Subscriptions & Refunds"]:::process

 B --> C4["■ Audit Logs (/admin/audit)"]:::admin
 C4 --> C4A["Immutable Timeline of All Staff Actions"]:::process
```

```

Operational Flow Breakdown:

- **Admin Authentication:** Guarded login at /admin/login redirects authorized staff to dark navy layout (/admin).
- **Verification Queue Review:** Inspect candidate selfie liveness and Aadhaar ID uploads; issue green verified badges.
- **Report Moderation:** Review flagged accounts (harassment, fake profiles, spam) and issue warnings or instant account bans.
- **Financial Oversight:** Audit total revenue, recent transactions, subscription renewals, and process refunds.
- **Audit Trail:** Immutable logging of every admin action with timestamp, admin ID, and IP address.